

Open-Set Single-Channel Blind Source Separation: A Per-Source SNR-Labeling Pitfall, a Deployment-Faithful Evaluation Protocol, and a Systematic Negative Result

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Abstract

Single-channel blind source separation (SC-BSS) of co-frequency communication signals is increasingly combined with per-source modulation identification, and a deployed receiver must additionally *reject* sources whose modulation it has never seen. Evaluating such open-set capability almost invariably involves conditioning detection metrics on the operating SNR. We show that this conditioning is fragile: a subtle per-source labeling convention—storing per-source SNR labels interleaved (`np.repeat`) while the corresponding score arrays are stacked slot-wise (`np.tile`)—misaligns every known-pool score with the wrong SNR bin and thereby *manufactures* a compelling but entirely spurious phenomenon: two out-of-distribution (OOD) scorer families appear strongly complementary across SNR, a training-free SNR-routed en-

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semble appears to lift the weighted-average AUROC from 0.50 to 0.625, and the effect appears multi-seed significant, robust to SIR/timing/carrier-offset perturbations, and gracefully degrading under SNR-estimation error. Every one of these properties is an artifact of the misalignment. We dissect the mechanism, explain why it survives exactly the checks practitioners trust, and show how it was caught—by sample-level alignment verification required to deploy a real blind SNR estimator. We then re-run the entire study under a corrected, deployment-faithful protocol (strict train/reference/test separation, unknown-free threshold calibration, verified labels) across six post-hoc OOD scorers, five seeds, SIR/timing/CFO perturbations, center-loss and embedding-dimension interventions, carrier-frequency separations, leave-one-modulation-out splits, a second attention-free backbone, and single-threshold operating points. The corrected finding is a systematic *negative* result: per-source detection of unknown modulations from separation embeddings is at chance everywhere (routed 0.498 ± 0.020 ; six-scorer oracle ceiling 0.549 ± 0.031 ; unknown detection rate 0.058 at a calibrated 5% false-rejection operating point). We attribute this to the separation objective, which shapes no known/unknown margin, and distill a protocol checklist for future open-set evaluations in signal processing.

Keywords: Blind source separation, Co-frequency communication, Open-set recognition, Out-of-distribution detection, Evaluation methodology, Data leakage, Complex-valued neural network

1. Introduction

Problem and motivation. Single-channel blind source separation (SC-BSS) recovers multiple unknown source signals from one observed mixture [1, 2]. For co-frequency communication signals, existing deep separators assume
5 a closed world: the modulation format of every source belongs to a known training alphabet. A deployed receiver (spectrum monitoring, cognitive radio, electronic warfare) faces transmitters whose waveforms it has never seen, so a practical system must answer two questions jointly: *what are the sources*, and *do I even know what they are*. The second question is an open-set recognition problem, and the natural toolkit—post-hoc out-of-distribution (OOD)
10 scoring of learned embeddings and logits [3, 4, 5, 6, 7]—is mature in computer vision and increasingly imported into signal processing.

Why evaluation is the hard part. Communication signals differ from images in one crucial respect: detection performance is meaningless without the
15 operating condition. Signal-to-noise ratio (SNR) is not merely a difficulty knob—classifier confidence, feature geometry, and every OOD score scale drift systematically with SNR [8, 9]. Practitioners therefore report *SNR-conditioned* metrics: per-SNR-bin AUROC profiles, and summary statistics that average bins weighted by their pair counts. This paper shows that
20 the conditioning step itself is a single point of failure. A one-line labeling convention—per-source SNR labels stored interleaved while the per-source scores they annotate are stacked slot-wise—misaligns every known-pool score with the wrong SNR. Because all OOD score scales drift with SNR, each mislabeled “per-SNR bin” then compares known and unknown pools at *different*
25 *ferent* true SNRs, and the pools separate by their SNR difference rather

than by any known/unknown structure. The result is a manufactured phenomenon with every surface property of a real discovery: it is consistent across five random seeds, “robust” to SIR, timing, and carrier-frequency perturbations, degrades “gracefully” under simulated SNR-estimation error, and
 30 yields a routing scheme with an apparently large, apparently significant gain (weighted-average AUROC 0.625 ± 0.031 vs. 0.50 pooled). All of it is an artifact.

What we do. We first dissect the pitfall: the exact repeat-vs-tile mechanism, the full catalogue of artifact results it produced, why it survives
 35 multi-seed averaging and every robustness check (it is *systematic*, not random), and how it was in fact caught—not by statistics, but by a sample-level alignment verification that a real blind SNR estimator forced us to write. We then rebuild the evaluation as a deployment-faithful protocol: strict train/reference/test separation, OOD reference statistics and rejection
 40 thresholds calibrated *without* any unknown-class data, verified per-source labels, and operating-point metrics (Det@ τ , FPR95, OSCR, joint accuracy) alongside AUROC. Under this protocol we re-run the complete study—six post-hoc scorers, five seeds, seven SNR bins, SIR/timing/CFO robustness, center-loss and embedding-dimension interventions, four carrier separations, estimated-SNR routing with a validated blind subspace SNR estimator, and
 45 leave-one-modulation-out splits.

The corrected finding is a clean negative result. Per-source OOD detection of unknown modulations from separation embeddings is at chance under every condition we tested: SNR-routed weighted-average AUROC $0.498 \pm$
 50 0.020 (was 0.625), post-hoc oracle over all six scorers 0.549 ± 0.031 (marginally

above chance, and inflated by per-bin max-selection), unknown detection rate 0.058 ± 0.038 at a calibrated 5% false-rejection operating point. The pooled AUROC ≈ 0.50 that the artifact had appeared to “explain away” was the honest number all along. We attribute the failure to the training objective: permutation-invariant waveform separation plus known-class cross-entropy
 55 shapes no margin against modulations it never sees, and neither capacity nor center-loss shaping changes this.

Contributions.

1. **Pitfall analysis.** A complete dissection of a per-source SNR-labeling
 60 pitfall in conditioned OOD evaluation: mechanism, the manufactured phenomenon it produces, an explanation of why it evades standard safeguards (multi-seed averaging, robustness checks, significance testing), and the alignment-verification step that catches it (Section 4).
2. **Protocol.** A corrected, deployment-faithful evaluation protocol for
 65 open-set SC-BSS: strict train/reference/test separation, unknown-free threshold calibration, verified label alignment, and operating-point metrics (Section 5).
3. **Systematic negative result with attribution.** Evidence that post-hoc per-source OOD detection of unknown modulations from separation
 70 embeddings is at chance across scorers, seeds, channel perturbations, interventions, and operating points, together with an analysis of *why*, and a protocol checklist for future work (Sections 6–7).

We present the pitfall as what it is: a cautionary analysis of a failure mode we identified and corrected in the course of this study, documented
 75 in the hope that the increasingly conditioned evaluation practice in signal

processing can avoid it. The remainder of the paper is organised as follows. Section 2 reviews the literature this work connects. Section 3 formalises the open-set SC-BSS problem, the backbone, and the OOD scorers. Section 4 analyses the pitfall; Section 5 the corrected protocol; Section 6 the corrected results. Section 7 discusses attribution and distills a checklist; Section 8 states limitations; Section 9 concludes.

2. Related Work

2.1. SC-BSS for communication signals

Blind separation of co-frequency communication signals from a single sensor violates the overdetermined requirement of classical ICA [11] and has driven a line of deep-learning approaches: convolutional separation networks [1], recurrent designs [12], attention-augmented models [13], complex-domain architectures [2], spatial-temporal fusion [14], and structured state-space backbones [15]. Blind separation has also been used as a front-end for co-channel modulation classification [16], but again under a closed-set label alphabet. All of these works evaluate under a closed-set protocol: the modulation alphabet of every source at test time is drawn from the training alphabet. The separator used as the backbone in this paper is a lightweight complex-valued CNN with complex squeeze-and-excitation (C-SE) channel attention (Section 3.2); the separator itself is not a contribution—the contribution is the evaluation analysis built on top of it.

2.2. Complex-valued neural networks

Complex-valued networks process I/Q waveforms in their native domain, implementing complex arithmetic via paired real operations [17, 18]. Channel

100 attention in the form of squeeze-and-excitation [19] transfers naturally to the complex setting; the backbone used here (Section 3.2) follows these design choices. The questions we ask are orthogonal to architecture: *do the learned features expose unknown modulations—and how would we know?*

2.3. Open-set recognition and OOD detection

105 Open-set recognition augments classification with a rejection option for unseen classes [20, 21]. Post-hoc OOD scorers require no retraining and are the family we evaluate: the maximum-softmax baseline [3], ODIN’s temperature-scaled, input-perturbed score [4], the energy score [5], class-conditional Mahalanobis distance [6], prototype (nearest-class-mean) distance, and a VOS-inspired virtual-outlier score [7] (used here in a strictly training-free, post-hoc sense—no outlier-aware loss is applied; see Section 3.3). Detection is typically measured by AUROC, with open-set-specific variants such as OSCR [10]. Embedding-shaping losses such as center loss [22] are a common training-side intervention, and recent analyses cau-
115 tion that strong closed-set features already determine much of open-set performance [23]. Standardised benchmarks [24] increasingly stress unified, leakage-free evaluation protocols—the concern at the centre of this paper.

2.4. Evaluation pitfalls and leakage

That subtle pipeline errors can manufacture publishable effects is documented across machine learning. Kapoor and Narayanan [25] catalogue leakage-induced overclaiming across hundreds of applied-ML papers and show that the affected results typically survive conventional robustness checks; Musgrave et al. [26] show that evaluation-convention inconsistencies,

not method improvements, account for much of the apparent progress in
 125 metric learning. The pitfall we analyse is of the same family—a structured-
 label misalignment—but with a twist that makes it especially dangerous
 in signal processing: it does not merely inflate a number, it manufactures
 a *condition-dependent structure* (an apparent SNR-dependent inversion of
 scorer rankings) that invites mechanistic storytelling and an obvious “ex-
 130 ploitation” (SNR routing), complete with a plausible robustness narrative.

2.5. SNR-dependent behaviour and blind SNR estimation

That classifier confidence and feature geometry degrade with noise is folk-
 lore in automatic modulation classification, where accuracy curves are rou-
 tinely reported as functions of SNR [8, 9]; conditioned evaluation is there-
 135 fore standard practice—and, per this paper, a place where label correct-
 ness deserves explicit verification. Our corrected protocol routes on an *esti-
 mated* SNR; classical blind SNR estimators (moment-based M2M4 and eigen-
 value/subspace methods) are surveyed in [27]. We find M2M4 inadequate for
 our signal class (two-source co-frequency mixtures with multipath) and vali-
 140 date a median-eigenvalue subspace estimator instead (Section 5.1).

3. Open-Set SC-BSS: System and Scorers

3.1. Problem setup

We observe a single complex baseband mixture of two co-frequency
 sources,

$$y(t) = \alpha_1 s_1(t) + \alpha_2 s_2(t) + n(t), \quad (1)$$

145 where each s_i is drawn from a modulation alphabet $m_i \in \mathcal{K} \cup \mathcal{U}$, with
 $\mathcal{K} = \{\text{BPSK, QPSK, 8PSK, 16QAM}\}$ the *known* classes seen in training

and $\mathcal{U} = \{64\text{QAM}, \pi/4\text{-DQPSK}, \text{MSK}, \text{OFDM-QPSK}\}$ held out. The system must recover $(\hat{s}_1, \hat{s}_2)$ and assign each source a label $\hat{m}_i \in \mathcal{K} \cup \{\text{unknown}\}$. Evaluation uses three protocols: **kk** (both sources known), **ku** (exactly one unknown), and **uu** (both unknown). For OOD evaluation, the *known pool* comprises both sources of the kk mixtures and the *unknown pool* comprises the OOD source of each ku mixture together with both sources of the uu mixtures.

3.2. OpenSetCSE architecture

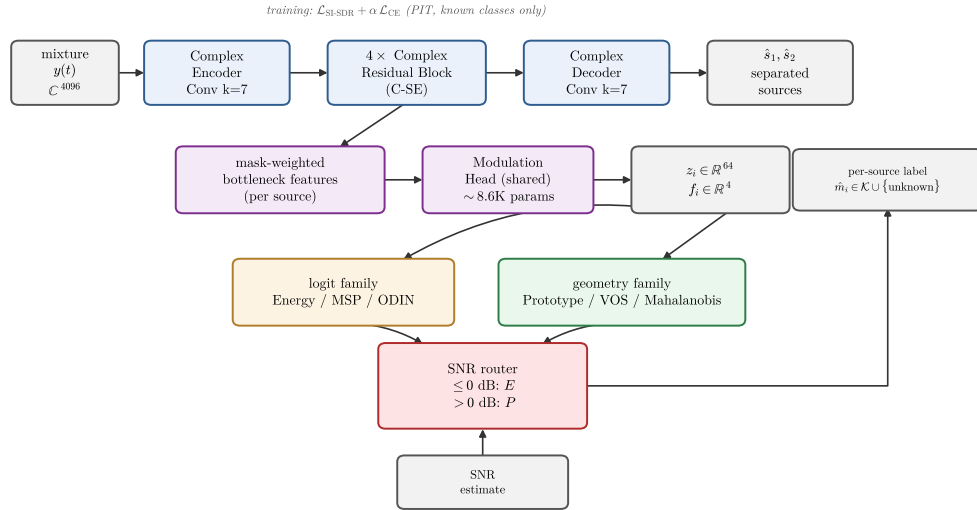


Figure 1: The OpenSetCSE system under study. Top: the C-SE separation backbone. Bottom: the shared per-source modulation head produces an embedding z_i and logits f_i ; the dashed SNR router (Energy at SNR ≤ 0 dB, Prototype otherwise) is the a-priori routing rule whose apparent effectiveness was manufactured by the labeling pitfall of Section 4 and is re-evaluated under the corrected protocol of Section 5.

155 The separation backbone is a lightweight complex-valued CNN with complex squeeze-and-excitation (C-SE) channel attention: a complex encoder (ComplexConv1d + complex batch normalisation), a stack of four complex residual blocks with C-SE, and a complex decoder producing two separated waveforms $\hat{s}_1, \hat{s}_2 \in \mathbb{C}^T$. On top of the backbone, a shared per-source *modu-*
160 *lation head* aggregates each output’s mask-weighted bottleneck features into a 64-dimensional embedding z_i and $K = 4$ logits f_i (one per known modulation). The head adds $\sim 8.6\text{K}$ parameters; the full model has $\sim 69\text{K}$ parameters. Training is end-to-end with a permutation-invariant multi-task loss,

$$\mathcal{L} = \mathcal{L}_{\text{SI-SDR}} + \alpha \mathcal{L}_{\text{CE}}, \quad (2)$$

165 where $\mathcal{L}_{\text{SI-SDR}}$ is the scale-invariant signal-to-distortion ratio (SI-SDR) loss under the PIT convention [28, 29] and $\mathcal{L}_{\text{CE}}$ is the cross-entropy of the per-source modulation labels under the same permutation; we use $\alpha = 1$. Only the four known modulations appear in training—no unknown-class data, synthetic or real, is used at any stage.

170 3.3. Per-source OOD scoring

Given per-source embeddings $z \in \mathbb{R}^{64}$ and logits $f \in \mathbb{R}^K$, we evaluate six inference-time scorers (larger = more OOD):

$$\text{Energy: } E(f) = -\log \sum_k \exp f_k, \quad (3)$$

$$\text{Prototype: } P(z) = \min_k \|z - \mu_k\|_2, \quad (4)$$

$$\text{VOS-inspired: } V(z) = \min_j \|z - v_j\|_2, \quad (5)$$

$$\text{Mahalanobis: } M(z) = \min_k (z - \mu_k)^\top \Sigma^{-1} (z - \mu_k), \quad (6)$$

$$\text{MSP: } S(f) = -\max_k \text{softmax}(f)_k, \quad (7)$$

$$\text{ODIN: } O(x) = -\max_k \text{softmax}(f(\tilde{x})/T)_k, \quad (8)$$

with class prototypes μ_k (and the shared covariance Σ) fit on a known-class reference pool, and virtual outliers v_j synthesised by extrapolation from the
 175 class prototypes in the spirit of [7]. We stress that Eq. (5) is *VOS-inspired*: a training-free, post-hoc heuristic that borrows the virtual-outlier construction of VOS but applies no outlier-aware training loss; it should not be read as the original VOS method. ODIN [4] uses temperature $T = 1000$ and a perturbation magnitude $\varepsilon = 5 \times 10^{-3}$ extended to the complex mixture as
 180 $\tilde{x} = x + \varepsilon (\text{sign } \Re g + j \text{sign } \Im g)$, $g = \nabla_x \log \text{softmax}(f/T)_{\hat{y}}$. Since no unknown-class data is available at validation time by construction, ε is fixed a priori rather than tuned. Under the corrected protocol (Section 5) the reference statistics (μ_k, Σ, v_j) are fit on a *disjoint* reference pool (seed 88888), never on test data.

185 4. The Pitfall: Repeat-vs-Tile Per-Source SNR Labels

4.1. Mechanism

The evaluation pipeline dumps, for every test sample, per-source OOD scores and the per-source SNR labels used to bin them. Two different flattening conventions met at this interface. The per-source score, embedding, and modulation-label arrays were stacked *tile-wise*: all slot-1 entries for every
 190 mixture, then all slot-2 entries,

$$\text{scores} = \text{concatenate}([\text{slot 1 of all mixtures}, \text{slot 2 of all mixtures}]), \quad (9)$$

while the per-source SNR labels were written *interleaved* (`np.repeat`): each mixture’s SNR twice in a row,

$$\text{snr_label} = \text{np.repeat}(\text{SNR}_{\text{mixture}}, 2). \quad (10)$$

Entry j of the score array is the $(j \bmod N)$ -th mixture’s slot $\lfloor j/N \rfloor + 1$ (true
 195 SNR $\text{SNR}_{j \bmod N}$), but it was paired with label $\text{SNR}_{\lfloor j/2 \rfloor}$. Because the test set is generated in SNR blocks (192 mixtures per SNR per protocol), the misalignment is not random but *structured*: stored “bin i ” of the known pool in fact contains scores from true bins $2i$ and $2i+1$, and—past the slot-1/slot-2 boundary—from true bins whose SNR differs from the stored label by up to
 200 15 dB (Fig. 2). The unknown pool, whose SNR array was masked with the same permutation-invariant-training swap masks as its scores, was always exact. Every “per-SNR” AUROC of the known pool therefore compared known sources at one set of true SNRs against unknown sources at *another*.

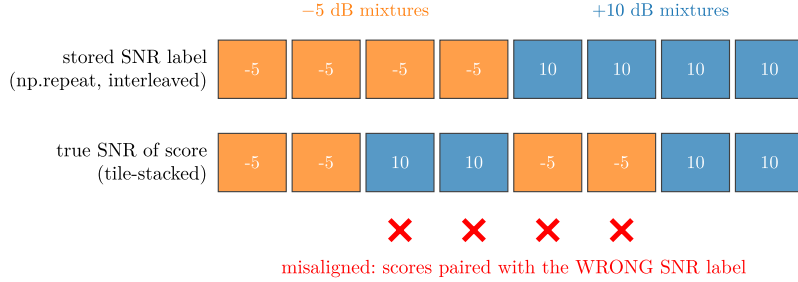


Figure 2: Repeat-vs-tile misalignment on a four-mixture toy example. The score array is stacked slot-wise (all slot-1, then all slot-2), so the true SNR sequence of its entries is $[-5, -5, 10, 10, -5, -5, 10, 10]$; the stored label sequence repeats each mixture’s SNR in place, $[-5, -5, -5, -5, 10, 10, 10, 10]$. Half of all entries (red crosses) are paired with the wrong SNR. In the real test set (1344 kk mixtures in seven SNR blocks) the same mechanism assigns every stored “SNR bin” of the known pool scores from other true bins, with label–truth mismatches of up to 15 dB.

A one-line check exposes the corruption. Binning the stored *modulation* labels of the known pool by the stored (repeat) SNR labels gives per-bin
 205 class counts 114/96/100/74—impossible, since the kk protocol contributes exactly 96 mixtures of each known modulation per true SNR bin; binning by the corrected (tile) labels recovers exactly 96/96/96/96.

4.2. The manufactured phenomenon

210 Why does the misalignment produce *structure* rather than mere noise? All OOD score scales drift with SNR: noise flattens logits and inflates embedding distances. A stored “bin” whose known entries actually come from *lower*-SNR blocks therefore contains known scores shifted toward the unknown end of the scale (and vice versa), so the per-bin AUROC measures
 215 the score’s *SNR separability* between the mismatched pools—a large, deter-

ministic signal. Because the Energy score and the Prototype score drift with SNR in opposite directions relative to the pool imbalance at each stored bin, the artifact presented as a perfect complementarity: Prototype/VOS-inspired apparently reached 0.84 ± 0.02 AUROC at 10 dB while collapsing below 0 dB, and Energy apparently peaked at 0.74 ± 0.03 at -5 dB while falling to 0.22 at 10 dB (Fig. 3a). The obvious exploitation—route each sample to the scorer that “works” at its SNR—then appeared to lift the pair-weighted average AUROC from 0.50 (pooled) to 0.625 ± 0.031 over five seeds, against a post-hoc oracle bound of 0.641 (three scorers) to 0.681 (six scorers).

The artifact did not stop at the headline. It produced a complete, internally consistent research narrative, every row of which dissolved under corrected labels (Table 1): simulated SNR-estimation noise of $\sigma = 1/3/6$ dB appeared to degrade the routed AUROC “gracefully” (0.597/0.593/0.576); SIR, timing-offset, and carrier-offset perturbations left it “robust” (0.58–0.66); carrier gaps of 10–500 Hz left it “unchanged” (0.615–0.633); a center-loss intervention appeared to *hurt* in an informative way (0.568); and an embedding-dimension ablation appeared to show capacity independence (0.588–0.631).

Table 1: The artifact catalogue: apparent results produced by the repeat-vs-tile misalignment vs. the same quantities recomputed under corrected labels (weighted-average per-SNR AUROC, 5 seeds unless noted). Full corrected numbers in Section 6.

Quantity	Artifact	Corrected
SNR-routed ensemble	0.625 ± 0.031	0.498 ± 0.020
Oracle, 3 scorers (post-hoc)	0.641	0.514 ± 0.013
Oracle, 6 scorers (post-hoc)	0.681	0.549 ± 0.031
SNR noise $\sigma = 1/3/6$ dB	0.597/0.593/0.576	0.502/0.502/0.501
SIR/timing/CFO (3 seeds)	0.58–0.66	0.487–0.494
Carrier gap 10–500 Hz	0.615–0.633	0.496–0.503
Center loss ($\lambda=0.1$)	0.568	0.487 ± 0.012
Embedding dim 16/32/128 (3 seeds)	0.631/0.588/0.618	0.508/0.477/0.500

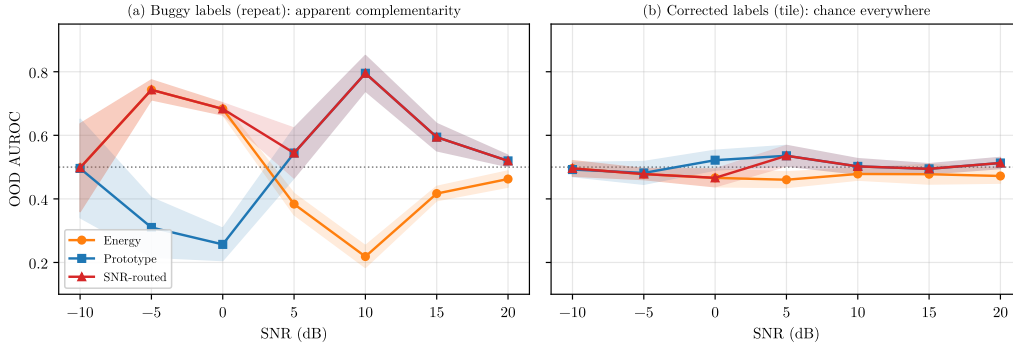


Figure 3: Per-SNR OOD AUROC of the *same* score dumps (5-seed mean $\pm$ std) under (a) the buggy repeat labels and (b) the corrected tile labels. Panel (a) shows the manufactured complementarity—Energy apparently peaking at low SNR, Prototype at high SNR, the routed rule apparently following the better branch. Panel (b) shows the same scores with verified labels: flat at chance at every SNR.

4.3. Why the artifact is seductive

Three properties make this failure mode dangerous, and we list them
235 because each corresponds to a safeguard that practitioners (including us)
trusted:

It is systematic, not random. The misalignment is a deterministic function of the data layout, identical for every seed, every model, every condition. Multi-seed averaging therefore does not dilute it—it *confirms* it, with a comfortably small standard deviation (± 0.031). Statistical significance testing
240 offers no protection either: a p -value computed on the mislabeled bins is significant precisely because the SNR-mismatch signal is real; what is wrong is the interpretation of the bins.

It survives every robustness check. SIR imbalance, timing offsets, and
245 carrier-frequency offsets perturb the signals, not the label geometry, so the artifact passes the exact perturbation battery reviewers ask for—and each pass reads as independent evidence. The simulated SNR-noise sweep is worse: “graceful degradation” under increasing σ is precisely what the *artifact* predicts, because noising the SNR only blurs the SNR-mismatch signal the bins
250 measure. The check most discriminating between truth and artifact would have been the one nobody thinks to run: verifying that the labels themselves are aligned.

It is mechanistically narratable. Each artifact number admitted a plausible post-hoc story (logit-based confidence “naturally” inverting at high SNR; geometry-based scores “naturally” requiring de-noised embeddings), which is
255 how the artifact survived internal scrutiny across two manuscript revisions. A sufficiently flexible intuition can explain any deterministic pattern; the lesson

is that a convincing mechanism story is not evidence of a correct pipeline.

4.4. How it was caught

260 The bug was not found by a statistical test but by an engineering necessity. Replacing the oracle SNR in the router with a *real blind SNR estimator* (Section 5.1) required estimating the SNR from each regenerated test waveform and pairing the estimate with the stored score of the same sample. The sample-level alignment verification written for that purpose—
265 regenerate the test set with its deterministic seed, then check the stored arrays against the data order element by element—failed immediately: the stored per-source SNR sequence matched `np.repeat` while the score array order matched `np.tile`. Recomputing every per-SNR quantity under corrected labels collapsed the routed AUROC to 0.498. We regard the verification step itself—*align labels against regenerated data before trusting any*
270 *conditioned metric*—as the portable lesson, and it is now part of the protocol in Section 5.

5. Corrected, Deployment-Faithful Evaluation Protocol

The corrected protocol changes *how things are measured*, not what is
275 trained. Its five rules:

P1. Verified label alignment. After any dump, per-source label arrays are verified against regenerated data: sequence equality of the per-mixture SNR order, pool sizes, per-bin sample counts, and per-bin modulation-label multisets (the 96/96/96/96 check of Section 4). Every number in Section 6
280 comes from dumps that pass all checks on all five seeds.

P2. Strict train/reference/test separation. Training uses its own stream; OOD reference statistics (μ_k, Σ, v_j) are fit on a *disjoint reference pool* (deterministic seed 88888) drawn under the test protocol; the test set uses seed 99999. No test sample enters any fitted quantity. (The artifact-era
 285 pipeline had already verified that re-fitting reference statistics on a held-out pool changed every AUROC by < 0.003 ; the corrected protocol makes the separation mandatory rather than optional.)

P3. Unknown-free threshold calibration. AUROC is threshold-free and can hide an unusable operating point. We therefore also fix a *single*
 290 *global rejection threshold* τ at the 95th percentile of the routed score on the reference-pool knowns (a 5% false-rejection-rate target), calibrated *without* any unknown-class data, and report the operational metrics: Det@ τ (unknown detection rate at τ), FRR@ τ , FPR95, OSCR [10], and joint accuracy (known accepted and correctly classified, plus unknown rejected, over all sam-
 295 ples). Because Energy and Prototype scores live on disjoint scales, the routed score is calibrated on the reference pool directly, with routed low-SNR reference samples handled by a provable below-threshold argument (score ranges verified per seed).

P4. A-priori routing decisions. The routing boundary (0 dB) is fixed
 300 a priori from training-range profiles, never tuned on test data; the post-hoc per-bin oracle is reported separately and interpreted as an upper bound subject to max-selection bias, not as an achievable number.

P5. Real blind SNR estimation for deployed routing. A deployed router cannot read the true SNR. We use a validated blind subspace es-
 305 timator (Section 5.1) and report routing both with the estimate and, for

diagnosis, with ground truth.

5.1. A validated blind SNR estimator

The estimator is waveform-only: the noise floor is read off as the median eigenvalue of the $L=64$ Toeplitz autocorrelation matrix of the received mixture, and the signal power follows by subtraction. Table 2 summarises its quality on all 3360 test mixtures (480 per SNR bin): per-bin bias between -0.57 and -0.61 dB at the extremes and within -0.24 dB mid-range, overall RMSE 0.51 dB (bias -0.31 dB, std 0.40 dB), uniformly across kk/ku/uu protocols (per-protocol RMSE within 0.17–0.92 dB per bin). The classical M2M4 moment estimator [27] is inadequate for this signal class (overall RMSE 13.99 dB): its kurtosis assumption breaks for two-source co-frequency mixtures with RRC pulse shaping and three-tap multipath—the mixture’s measured fourth moment is dominated by the cross-term of the two sources and by the channel-induced constellation spread rather than by the single-source modulation kurtosis the estimator is calibrated on, so its SNR read-out saturates and compresses (Fig. 4). Equivalent Gaussian-noise calibration places the subspace estimator at $\sigma \approx 0.51$ dB.

Table 2: Blind SNR-estimation quality (all 3360 test mixtures, 480 per bin). Subspace = median-eigenvalue of the Toeplitz autocorrelation; M2M4 shown for contrast.

True SNR (dB)	Bias (dB)	Std (dB)	RMSE (dB)
−10	−0.57	0.59	0.82
−5	−0.24	0.26	0.35
0	−0.14	0.15	0.20
+5	−0.14	0.12	0.18
+10	−0.17	0.17	0.24
+15	−0.30	0.22	0.37
+20	−0.61	0.59	0.85
Overall	−0.31	0.40	0.51
M2M4 overall	−8.71	10.95	13.99

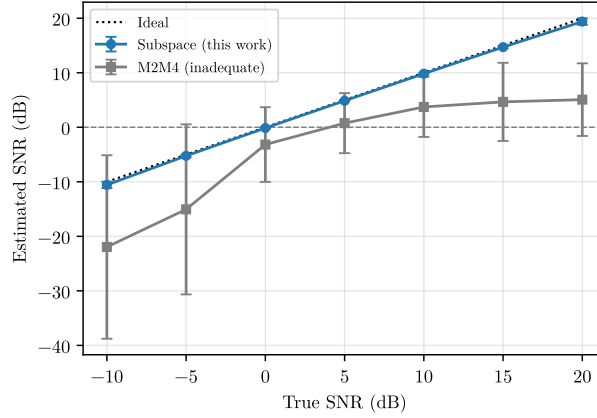


Figure 4: True vs. estimated SNR (mean $\pm$ std over 480 mixtures per bin). The subspace estimator tracks the ideal line across the full $[-10, 20]$ dB range; M2M4 saturates and is unusable for this signal class.

6. Corrected Results: A Systematic Negative Result

6.1. Setup

325 *Data.* All signals are synthetic complex baseband waveforms: linear modulations (and OFDM-QPSK) with root-raised-cosine pulse shaping, a three-tap multipath fading channel, and additive white Gaussian noise [8]. Each mixture contains two sources with carrier frequencies 2000 Hz and 2005 Hz (each with an independent ± 5 Hz jitter) at $T = 4096$ samples and
330 a 16 kHz sample rate. Training SNR is uniform in $[-5, 20]$ dB; test SNR spans $\{-10, -5, 0, 5, 10, 15, 20\}$ dB, i.e. including SNRs *below* the training range. Test sets are deterministic (seed 99999) with 200 mixtures per SNR for the kk/ku protocols and 100 for uu; the reference pool uses seed 88888. No external dataset is required; the generator is released with the code.

335 *Training.* OpenSetCSE (~ 69 K parameters), 100 epochs, batch size 16, Adam at 10^{-3} , loss of Eq. (2) with $\alpha = 1$. Five seeds (42–46). *Every* reported number is multi-seed—necessary, though this paper demonstrates at some length that it is not *sufficient*.

6.2. Closed-set sanity

340 Table 3 shows that the backbone itself works and is unaffected by the pitfall (the closed-set path builds its masks consistently): SI-SDR turns positive at $\text{SNR} \geq 10$ dB and closed-set classification accuracy is 0.41 (chance 0.25). The difficulty addressed in this paper is genuinely in OOD detection, not separation. Pooled (all-SNR) OOD AUROCs are likewise unaffected by
345 the pitfall—they use no SNR labels—and were honest all along (Table 4).

Table 3: Closed-set (kk) performance, 5-seed mean $\pm$ std (unaffected by the pitfall).

Metric	Value
Pooled SI-SDR (dB)	-0.88 ± 0.09
SI-SDR at 10 dB (dB)	$+0.47 \pm 0.14$
SI-SDR at 20 dB (dB)	$+0.67 \pm 0.16$
Classification accuracy	0.407 ± 0.030

Table 4: Pooled OOD AUROC (all SNRs mixed), 5-seed mean $\pm$ std (unaffected by the pitfall). All scorers are near chance.

Scorer	Pooled AUROC
Energy	0.482 ± 0.018
Prototype	0.509 ± 0.013
VOS-inspired	0.509 ± 0.013
Mahalanobis	0.524 ± 0.046
MSP	0.433 ± 0.010
ODIN	0.486 ± 0.016

6.3. Corrected per-SNR profiles: flat everywhere

Fig. 3b already showed the corrected Energy/Prototype profiles; Fig. 5 extends this to all six scorers. Every scorer sits within seed noise of chance at every SNR (per-bin range roughly 0.42–0.60, all within ~ 2 standard errors of 0.5); no scorer family works at any SNR, and there is no complementarity left to exploit. Table 5 quantifies the summary: the a-priori SNR-routed ensemble reaches 0.498 ± 0.020 —exactly chance. The post-hoc oracle over the

three primary scorers is 0.514 ± 0.013 , and over all six scorers 0.549 ± 0.031 . The six-scorer oracle deserves a caveat: it takes a per-bin *maximum* over six
355 scorers after seeing the data, so it is biased above 0.5 by construction (max-selection on seven bins), and even so it reaches only 0.55. The deployable, a-priori quantities—single scorers and the routed rule—are at chance. Mahalanobis, the strongest single scorer (0.521 ± 0.050 weighted, 0.524 ± 0.046 pooled), owes its margin largely to one seed and stays within seed noise of
360 chance.

Table 5: Corrected main result: weighted-average per-SNR AUROC, 5-seed mean $\pm$ std (pair-weighted bins, verified labels). Oracle = post-hoc best scorer per bin (upper bound, max-selection biased).

Scorer	Weighted-avg AUROC
Energy	0.476 ± 0.019
Prototype	0.506 ± 0.010
VOS-inspired	0.505 ± 0.010
Mahalanobis	0.521 ± 0.050
MSP	0.431 ± 0.009
ODIN	0.480 ± 0.019
SNR-routed (a-priori rule)	0.498 ± 0.020
Oracle, 3 scorers (post-hoc)	0.514 ± 0.013
Oracle, 6 scorers (post-hoc)	0.549 ± 0.031

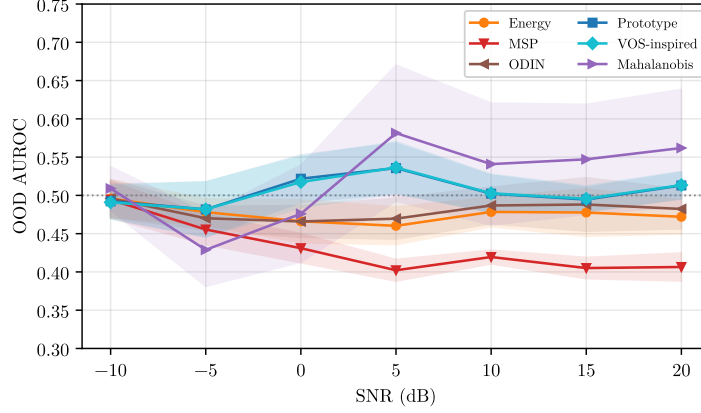


Figure 5: Corrected per-SNR OOD AUROC of all six scorers (5-seed mean $\pm$ std). Every profile is flat at chance; the artifact’s apparent complementarity (Fig. 3a) is gone.

6.4. Robustness: flat under every perturbation

The artifact’s robustness narrative inverts: under corrected labels there is nothing left to degrade. Simulated SNR-estimation noise of $\sigma = 1/3/6$ dB leaves the routed AUROC at 0.502/0.502/0.501 (± 0.014 –0.018; 50 Monte-
 365 Carlo trials per seed)—flat, because there is no routing gain to lose. The same holds under the physical perturbations of Table 6: SIR from -6 to $+6$ dB, random symbol-timing offsets, and a 25 Hz carrier jitter all leave the routed AUROC within 0.487–0.494 (3 seeds). Routing with the *estimated* SNR from the subspace estimator of Section 5.1 gives 0.501 ± 0.017 on corrected
 370 true-SNR bins (fully deployed, estimated-SNR bins: 0.471 ± 0.014 ; M2M4: 0.444)—again chance, confirming that the estimator quality is no longer a bottleneck because there is no gain for it to bottleneck.

Table 6: Corrected routed weighted-avg AUROC under channel perturbations (3 seeds, mean $\pm$ std; ground-truth SNR routing, a-priori 0 dB boundary).

Condition	Routed AUROC
Baseline	0.487 ± 0.014
SIR -6 dB	0.494 ± 0.008
SIR -3 dB	0.490 ± 0.007
SIR 0 dB	0.490 ± 0.012
SIR $+3$ dB	0.490 ± 0.011
SIR $+6$ dB	0.492 ± 0.009
Timing offset	0.492 ± 0.009
CFO jitter 25 Hz	0.490 ± 0.011

6.5. Interventions: training side, capacity, carrier gap

Center loss ($\lambda=0.1$, 5 seeds) [22]: routed 0.487 ± 0.012 , oracle 0.502 ± 0.009 —no gain. The artifact-era reading (“center loss weakens the high-SNR prototype signal the router relies on”) was a story about a signal that did not exist; the corrected reading is simpler: within-class compactness shaping cannot create a margin against classes never seen in training.

Embedding dimension (16/32/64/128, 3 seeds): routed 0.508, 0.477, 0.498, 0.500; oracles 0.530, 0.505, 0.514, 0.517—flat and non-monotone. OOD performance is not capacity-limited.

Carrier-frequency separation (10/50/100/500 Hz, model trained at 5 Hz, 5 seeds): routed $0.500 \pm 0.022/0.503 \pm 0.020/0.496 \pm 0.018/0.503 \pm 0.011$. The negative result is not an artifact of the training carrier gap.

385 6.6. Single-threshold operating point

AUROC can hide an unusable operating point; the deployment question is what happens at one calibrated threshold. Under protocol rule P3 (reference-pool calibration, 5% FRR target, no unknown-class data), Table 7 and Fig. 6 give the operational truth: the routed detector rejects only 0.058 ± 0.038 of unknown sources at its calibrated threshold, with OSCR 0.197 ± 0.012 and joint known/unknown accuracy 0.217 ± 0.028 . The per-SNR breakdown shows the detection rate is *exactly zero* at and below 0 dB and reaches only 0.15 even at 20 dB: at a single, honestly calibrated operating point, the system detects essentially no unknown sources anywhere. A rank-normalised
395 variant (each scorer mapped through its known-pool empirical CDF before routing, so a single threshold is scale-fair) changes the picture only marginally (Det 0.070 ± 0.031).

Table 7: Single-threshold operating point (protocol rule P3; threshold τ = 95th percentile of reference-pool known scores, 5% FRR target; 5 seeds, mean $\pm$ std). Det@ τ = unknown detection rate.

Method	Det@ τ	FRR@ τ	FPR95	OSCR	JointAcc
Routed	0.058 ± 0.038	0.051 ± 0.003	0.954 ± 0.010	0.197 ± 0.012	0.217 ± 0.028
Prototype	0.058 ± 0.038	0.051 ± 0.003	0.947 ± 0.013	0.190 ± 0.013	0.217 ± 0.028
VOS-inspired	0.058 ± 0.037	0.051 ± 0.003	0.947 ± 0.012	0.190 ± 0.012	0.217 ± 0.027

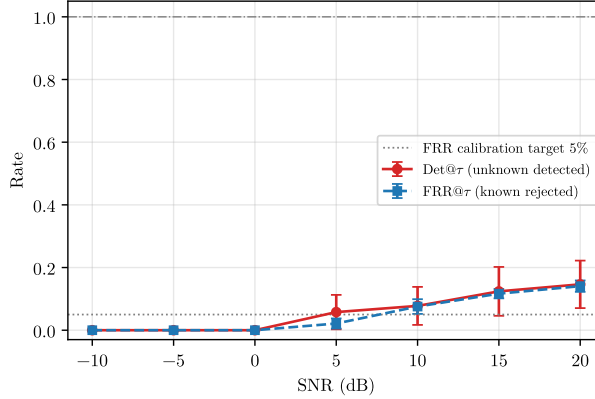


Figure 6: Per-SNR operating point of the routed detector at the calibrated threshold (variant of Table 7): unknown detection rate $\text{Det@}\tau$ and known false-rejection rate $\text{FRR@}\tau$. Detection is exactly 0 at $\text{SNR} \leq 0$ dB and reaches only 0.15 at 20 dB—the detector is operationally blind to unknown sources at a 5% FRR budget.

6.7. Leave-one-modulation-out and cross-backbone checks

Leave-one-modulation-out (LOMO). Each known modulation in turn is held out of training (3-mod known alphabet) and added to the unknown pool at test time; the routing boundary is selected on a pseudo-OOD *validation* split (seed 777777, the held-out known class playing the unknown role—never touching test data), i.e. the boundary is chosen *legally*, with no test unknowns and no oracle SNR profiling. Table 8 gives the full 4×3 grid. Two observations, both consistent with the rest of this paper. First, the legally selected boundary is *unstable*: 9 of 12 runs select 0 dB, but 3 of 12 find no usable crossover and fall back to an all-Prototype route (20 dB)—unstable precisely because, per Section 6.3, there is no real crossover to find. Second, the routed AUROC stays at chance (0.468–0.516) in every split and every seed, matching the corrected baseline 0.498; holding out one known

modulation at a time does not create detectability either.

Table 8: LOMO: routed weighted-avg AUROC (test seed 99999; unknown pool = held-out known modulation + the four real unknowns; routing boundary selected on pseudo-OOD validation seed 77777 only). “Boundary” is the validation-selected SNR switch; 20 dB = no crossover found, an all-Prototype route.

Held-out	Seed 42	Seed 43	Seed 44	Mean $\pm$ std
QPSK	0 dB, 0.516	0 dB, 0.505	20 dB, 0.501	0.508 ± 0.008
8PSK	0 dB, 0.514	0 dB, 0.506	20 dB, 0.506	0.509 ± 0.005
16QAM	0 dB, 0.481	0 dB, 0.493	0 dB, 0.472	0.482 ± 0.011
BPSK	0 dB, 0.498	0 dB, 0.468	20 dB, 0.491	0.486 ± 0.016

Cross-backbone (no-SE ablation). To check whether the negative result is specific to the C-SE backbone, we repeated the full pipeline with the SE blocks removed (`-no_se`, 3 seeds, 60.8K parameters). The no-SE variant is strictly weaker separator (pooled SI-SDR -1.15 dB vs. -0.88 dB; closed-set
415 accuracy 0.356 vs. 0.407), yet the corrected evaluation looks the same: routed weighted-avg AUROC 0.508 ± 0.011 , single scorers 0.491 ± 0.031 (Energy), 0.531 ± 0.012 (Prototype), 0.532 ± 0.011 (VOS-inspired), and a three-scorer oracle of 0.536 ± 0.014 . Prototype sits marginally above chance (≈ 0.53), but
420 its per-SNR profile shows no low/high-SNR complementarity with Energy (Energy is flat at 0.47 – 0.50 across SNR), so there is again nothing for a router to exploit. The scorer-inversion phenomenon appears in *neither* backbone once the labels are correct: it was a property of the mislabeled evaluation, not of any feature representation.

425 7. Discussion: Attribution and Lessons

7.1. *Why is detection at chance?*

The negative result is over-determined by the controls, and the attribution points at the training objective rather than the readout.

The objective shapes no known/unknown margin. Eq. (2) asks the net-
 430 work to (i) reconstruct waveforms permutation-invariantly and (ii) classify the four known modulations. Nothing in it references out-of-alphabet signals: the network is never shown one, never penalised for mapping one close to a known prototype, and—because waveform reconstruction at this difficulty level is largely modulation-agnostic (RRC pulses through a three-tap
 435 channel carrying 2–16 point constellations)—never even pressured to encode modulation beyond what the four-way head needs. The unknown modulations are moreover *same-family neighbours* of the known ones (64QAM vs. 16QAM; $\pi/4$ -DQPSK vs. PSKs; MSK vs. BPSK-like CPM), so their embeddings land inside the known support at every SNR.

The controls exclude the alternatives. If the failure were readout capacity,
 440 widening the embedding would help—it does not (0.477–0.508 over 16–128 dims). If it were within-class scatter, center loss would help—it does not (0.487). If it were a property of one scorer family, some member of the six would separate somewhere—none does, and the post-hoc ceiling over all six
 445 is 0.549, max-selection bias included. If it were the carrier separation or channel conditions, the gap/perturbation sweeps would move it—they do not. The remaining explanation is the simplest: the features contain no usable known/unknown signal to begin with.

Consequence for practice. An open-set SC-BSS receiver cannot expect

450 post-hoc OOD scoring of separation embeddings to flag unknown modulations; the capability, if wanted, must be *built in* at training time (outlier exposure, margin-shaping losses, explicit abstention heads) or sourced from modulation-specific signal structure rather than generic embeddings. Whether such training-side mechanisms help on this benchmark is an open
 455 question the corrected protocol is designed to answer honestly.

7.2. A protocol checklist for conditioned OOD evaluation

We distill what would have caught the pitfall early, and what the corrected study needed. None of these is expensive; all are cheap relative to a manuscript built on an artifact.

- 460 1. **Align labels against regenerated data.** Whenever a metric is conditioned on a per-sample variable, regenerate the deterministic test set and verify every label array against the data order (sequence equality, per-bin counts, per-bin label multisets). This single step catches the entire repeat/tile family.
- 465 2. **Be suspicious of flattering structure.** A dramatic, mechanistically convenient pattern (scorer rankings inverting on cue) warrants pipeline verification *before* narrative-building, not after.
3. **Multi-seed averaging is not a pipeline check.** It averages away random error and confirms systematic error. Statistical significance
 470 likewise says nothing about label correctness.
4. **Separate train / reference / test strictly.** Reference statistics and thresholds fitted on test data are leakage even when their measured influence is small; make the separation structural (distinct deterministic seeds), not cosmetic.

- 475 5. **Calibrate operating points without unknown-class data.** Report $\text{Det}@_\tau$ at an unknown-free threshold alongside AUROC; a scorer can look acceptable in AUROC while rejecting nothing at any deployable threshold (here: $\text{Det} = 0.058$).
- 480 6. **Treat oracle bounds as bounds.** Per-bin post-hoc maxima over S scorers are biased upward by selection; report them with the deployable a-priori rule, never as the headline.
- 485 7. **Use real estimators for operating-condition variables.** An oracle-SNR router can hide a large deployment gap; validating the estimator (Table 2) forced the alignment check that exposed the pitfall.

8. Limitations

The negative result is bounded in scope. Data are synthetic (RRC pulse shaping, three-tap multipath, AWGN); real-world impairments could move the picture in either direction. Scenarios are two-source ($K=2$) with symbol-aligned sources and a modest known alphabet. The evidence covers one separator family (C-SE CNN, $\sim 69\text{K}$ parameters, plus its no-SE ablation)—and
 490 post-hoc scorers only: training-side open-set mechanisms (outlier exposure, VOS-style losses on synthetic outliers, abstention heads) are explicitly out of scope and remain the most plausible way to change the picture. Finally, the
 495 SNR estimator is validated on the same synthetic signal class it is used on.

9. Conclusion

We set out to evaluate open-set per-source modulation detection on top of a single-channel blind separator, and found two things. First, a cautionary one: a one-line repeat-vs-tile mismatch between per-source SNR labels and per-source score arrays manufactured an entire apparent phenomenon—
500 SNR-dependent scorer complementarity, a 0.625 routed AUROC, graceful degradation, perturbation robustness—that survived multi-seed averaging, significance intuition, and every robustness check, and fell only to sample-level label alignment verification. Second, a clean negative one: under a corrected, deployment-faithful protocol, post-hoc OOD detection of unknown
505 modulations from separation embeddings is at chance across six scorers, five seeds, channel perturbations, training-side and capacity interventions, carrier separations, leave-one-modulation-out splits, a second attention-free backbone, and single-threshold operating points, because the separation objective shapes no known/unknown margin. We release the corrected pipeline and the archived artifact dumps together, and hope both the pitfall analysis and the protocol checklist prove useful to the growing literature on open-set evaluation in signal processing.

CRedit authorship contribution statement

515 **Bin Nong:** Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Weihong Fu:** Supervision, Project administration, Writing – review & editing. **Zhuoyun Jiang:** Resources, Validation, Writing – review & editing.

520 Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

525 Code, synthetic data generators, the corrected evaluation protocol, and both the corrected and the archived (artifact-laden) score dumps are available at https://github.com/BinNong/single_channel_source_seperation/tree/main/paper3_open_set. No external datasets are required.

Funding

530 This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Declaration of generative AI and AI-assisted technologies in the writing process

535 During the preparation of this work the authors used Claude Code (Anthropic) in order to assist with drafting, editing, and polishing the manuscript text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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